



Line Follower Robot Using PID Algorithm

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A line follower is a simple robot that follows a thick line drawn on the floor using infrared (IR) or some other optical sensors. This

line follower robot uses two motors with wheels on the rear and a castor wheel as support on the front.

Power is drawn from a battery fitted on the line follower. A logic system (usually a microcontroller) is used to control the movement of the robot by reading the line through sensors. The author's prototype is shown in Fig. 1.

The accuracy of the movement of a line follower mostly

depends on the motor movement and sensing system. Many algorithms are used to coordinate the line data read by the sensors to control the motors. But a simple and most precise control is obtained by PID (proportional integral derivative) algorithm.

The PID function formula having a combination of three successive terms [P] + [I] + [D] is as follows:

$$f(t) = [K_p \cdot e(t)] + [K_i \cdot e(t) \cdot d(t)] + [K_d \cdot de/dt]$$

where $e(t)$ is error value, which is the difference of required value to actual value,

K_p = constant for proportion

K_i = constant for integral

K_d = constant for derivative

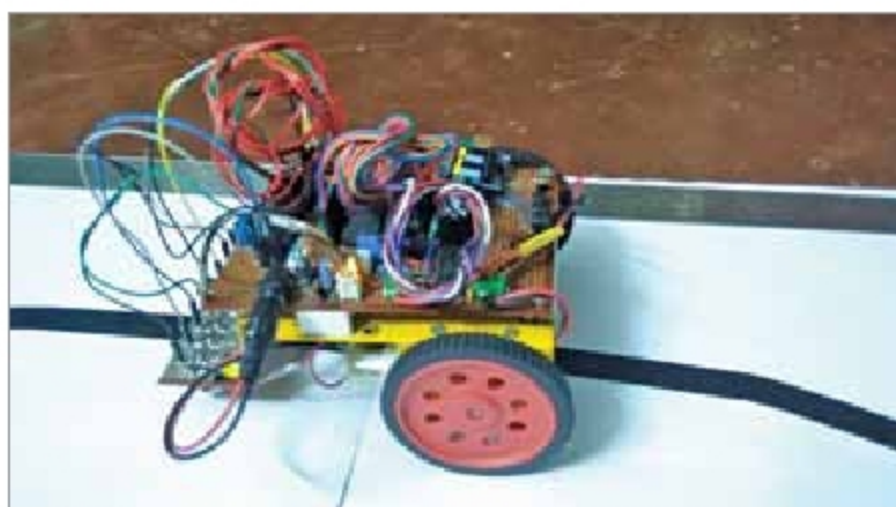


Fig. 1: Author's prototype

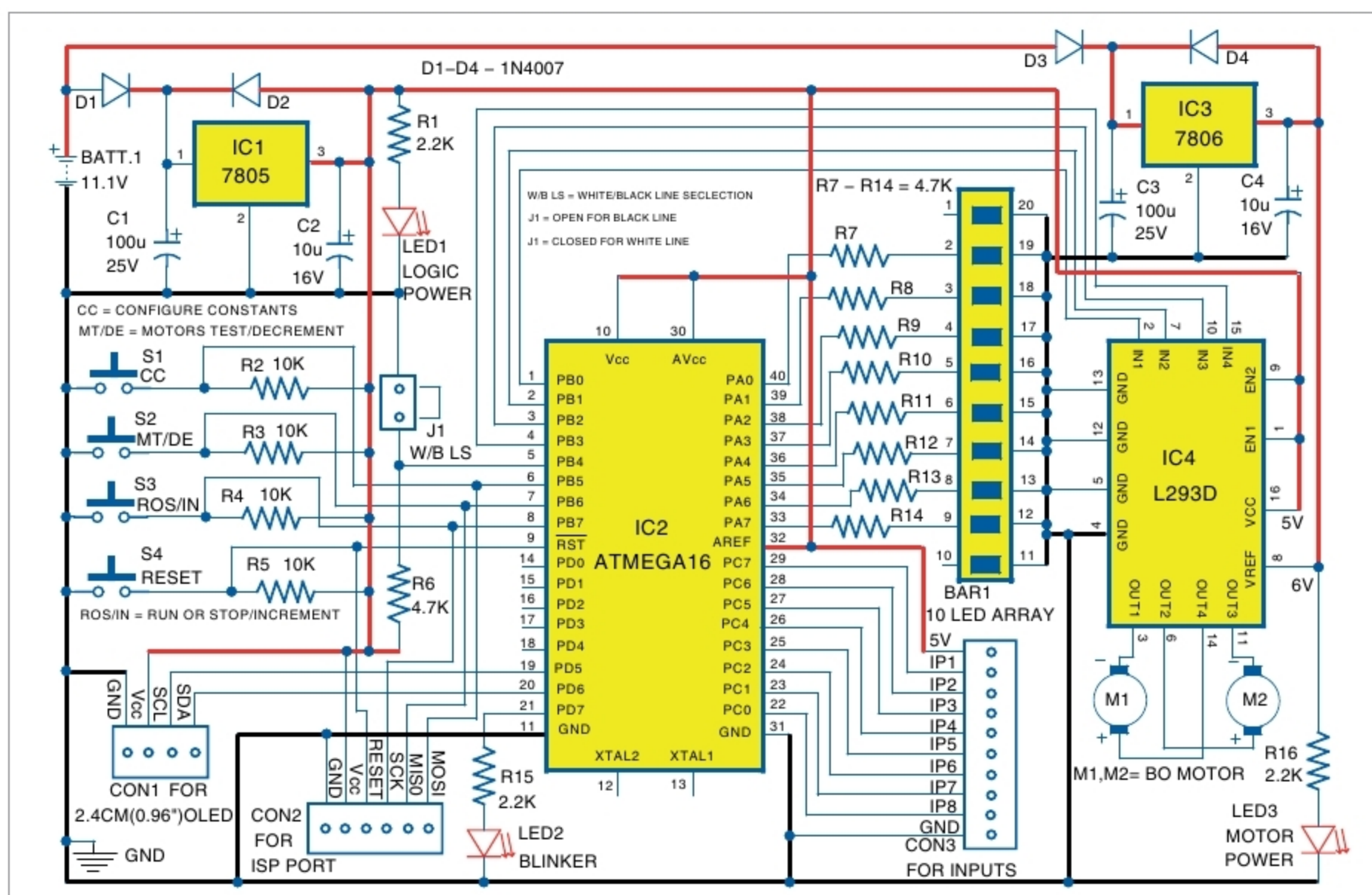


Fig. 2: Main circuit diagram of the custom PID line follower